IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

RoShawn Frazier
October 12, 2026

IBM Data Science Professional Certificate · Applied Data Science Capstone

GitHub: https://github.com/RoShawn-F/spacex-falcon9-landing-prediction



Executive Summary

Summary of methodologies

- Data collection: SpaceX REST API and web scraping of Wikipedia (90 Falcon 9 launches, 2010–2020)
- Data wrangling: created a landing label, Class (1 = landed, 0 = did not)
- EDA with SQL (10 queries) and with seaborn/matplotlib charts
- Interactive analytics: Folium maps and a Plotly Dash dashboard
- Predictive analysis: Logistic Regression, SVM, Decision Tree and KNN tuned with GridSearchCV (cv=10)

Summary of all results

- Landing success rose from 0% (2010–2013) to 90% (2019)
- KSC LC-39A has the best success rate: 10 of 13 launches landed
- SSO and VLEO orbits land reliably; GTO is the hardest (52%)
- Best model: Decision Tree (88.9% validation, 83.3% test accuracy); all 4 classifiers reach 83.3% on the test set
- SQL: 4 launch sites; first ground-pad landing 2015-12-22; 100 of 101 missions succeeded
- Maps and dashboard: sites are coastal and far from cities; KSC LC-39A holds 41.7% of all successful landings

90

Falcon 9 launches analyzed

66.7%

first stages landed overall

88.9%

best validation accuracy (Decision Tree)

$100M+

price gap per launch ($62M vs $165M+)

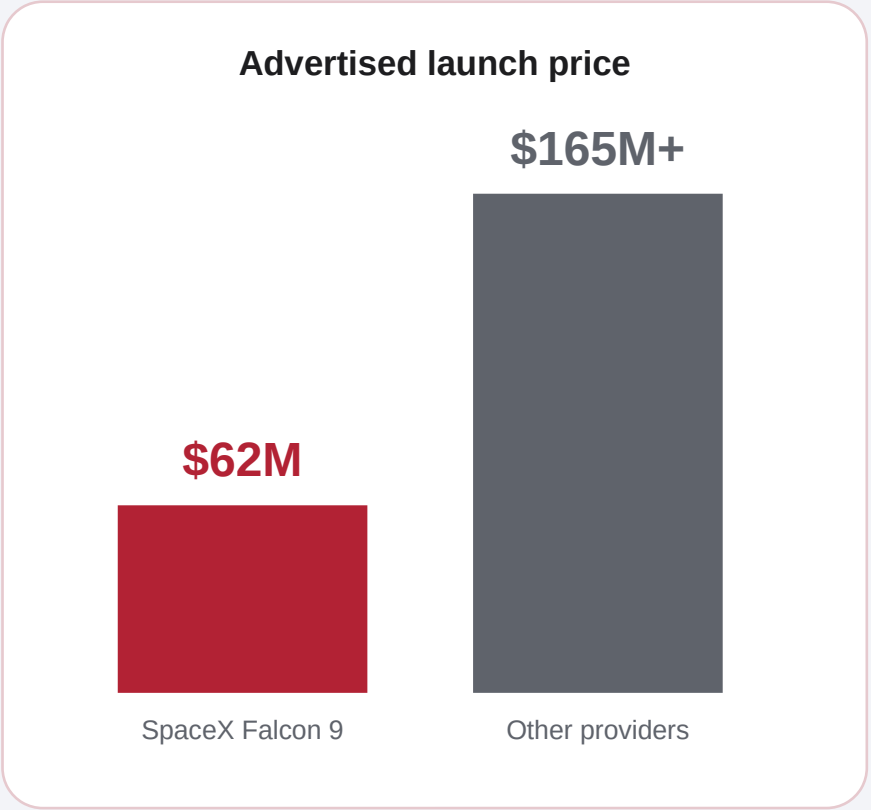
Introduction

Project background and context

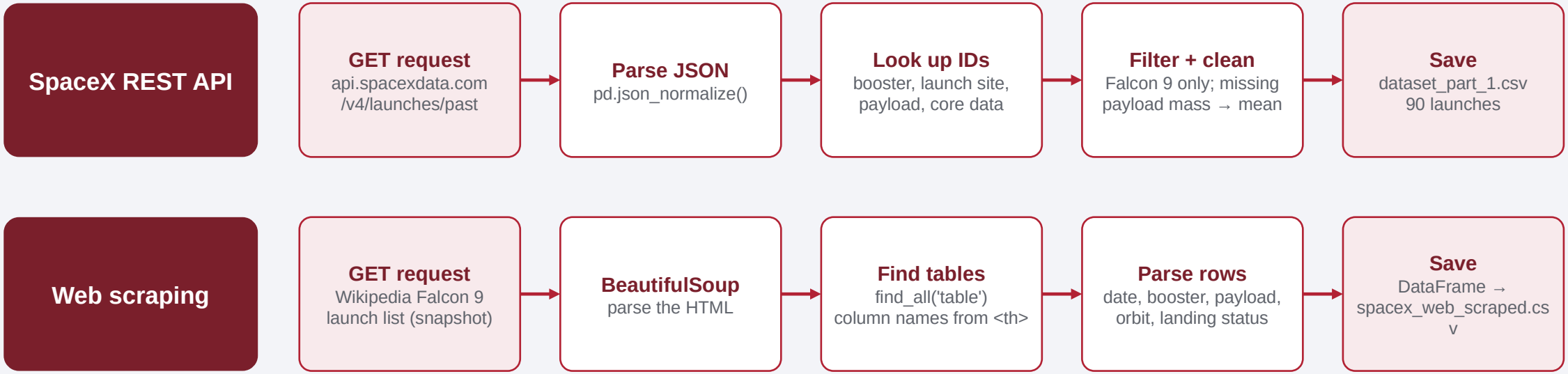
- SpaceX advertises Falcon 9 launches at about $62 million; other providers charge $165 million or more.
- Most of the savings comes from landing and reusing the first stage.
- Predicting whether the first stage will land lets a competitor (Space Y) estimate the true cost of a launch and bid against SpaceX.

Problems we want to answer

- Which factors (payload, orbit, launch site, booster version, experience) drive a successful landing?
- How have landing outcomes changed over time and across launch sites?
- Can a machine learning model predict a successful landing, and how accurately?



Methodology: Data Collection (SpaceX API and Web Scraping)



- Key phrases: requests.get() · .json() · json_normalize · helper functions turn IDs into readable values · BeautifulSoup · dictionary → DataFrame → CSV
- When the live SpaceX API stopped returning JSON (JSONDecodeError), the project switched to IBM's saved copy of the same API response, so the pipeline kept running.

GitHub (SpaceX API): https://github.com/RoShawn-F/spacex-falcon9-landing-prediction/blob/main/jupyter-labs-spacex-data-collection-api.ipynb

GitHub (web scraping): https://github.com/RoShawn-F/spacex-falcon9-landing-prediction/blob/main/jupyter-labs-webscraping.ipynb

Methodology: Data Wrangling



Outcome	Meaning	Count	Class
`True ASDS`	Landed on drone ship	41	1
`None None`	No landing attempt	19	0
`True RTLS`	Landed on ground pad	14	1
`False ASDS`	Failed on drone ship	6	0
`True Ocean`	Controlled ocean landing	5	1
`False Ocean`	Failed ocean landing	2	0
`None ASDS`	No attempt (drone ship)	2	0
`False RTLS`	Failed on ground pad	1	0

Results

- 60 of 90 launches landed: 66.7% success rate
- Launches by site: CCAFS SLC 40 (55), KSC LC 39A (22), VAFB SLC 4E (13)
- GTO is the most common orbit (27 launches)

GitHub (data wrangling): https://github.com/RoShawn-F/spacex-falcon9-landing-prediction/blob/main/labs-jupyter-spacex-Data_wrangling-v2.ipynb

Methodology: EDA with Data Visualization and SQL

Charts plotted and why

Chart	Why
Flight number vs. launch site (scatter)	Does experience improve landings?
Payload vs. launch site (scatter)	Do heavier payloads affect landing?
Success rate by orbit (bar)	Which orbits land most often?
Flight number vs. orbit (scatter)	Experience effect within each orbit
Payload vs. orbit (scatter)	Payload effect within each orbit
Yearly success trend (line)	How success changed over time

Feature engineering: one-hot encoded Orbit, LaunchSite, LandingPad and Serial (get_dummies) → 83 numeric features for modeling.

SQL queries performed (SQLite, %sql magic)

- Unique launch site names (DISTINCT)
- 5 records where the site begins with 'CCA' (LIKE)
- Total payload for NASA (CRS) (SUM)
- Average payload for booster F9 v1.1 (AVG)
- First successful ground-pad landing (MIN)
- Drone-ship successes with payload 4,000–6,000 kg
- Successful vs. failed mission outcomes (COUNT, GROUP BY)
- Boosters with the maximum payload (subquery, MAX)
- 2015 drone-ship failures by month (substr)
- Rank of landing outcomes, 2010-06-04 to 2017-03-20 (ORDER BY DESC)

GitHub (EDA visualization): https://github.com/RoShawn-F/spacex-falcon9-landing-prediction/blob/main/jupyter-labs-eda-dataviz-v2.ipynb

GitHub (EDA SQL): https://github.com/RoShawn-F/spacex-falcon9-landing-prediction/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Methodology: Interactive Visual Analytics (Folium and Plotly Dash)

Folium map objects

- Circle + text Marker for each launch site: shows where the 4 sites are
- MarkerCluster with green (landed) / red (failed) markers: shows each site's success rate
- MousePosition: reads coordinates to find the nearest coastline, railway, highway and city
- PolyLine + distance label: measures how close each landmark is (VAFB SLC-4E)

Plotly Dash dashboard

- Launch-site dropdown: compare all sites or one site
- Success pie chart: share of successes per site, or one site's success vs. failure
- Payload range slider (0–10,000 kg): focus on a payload range
- Payload vs. outcome scatter, colored by booster version: shows which payloads and boosters land
- Callbacks link the dropdown and slider to both charts

GitHub (Folium): https://github.com/RoShawn-F/spacex-falcon9-landing-prediction/blob/main/lab-jupyter-launch-site-location-v2.ipynb

GitHub (Plotly Dash app): https://github.com/RoShawn-F/spacex-falcon9-landing-prediction/blob/main/spacex_dash_app.py

Methodology: Predictive Analysis (Classification)



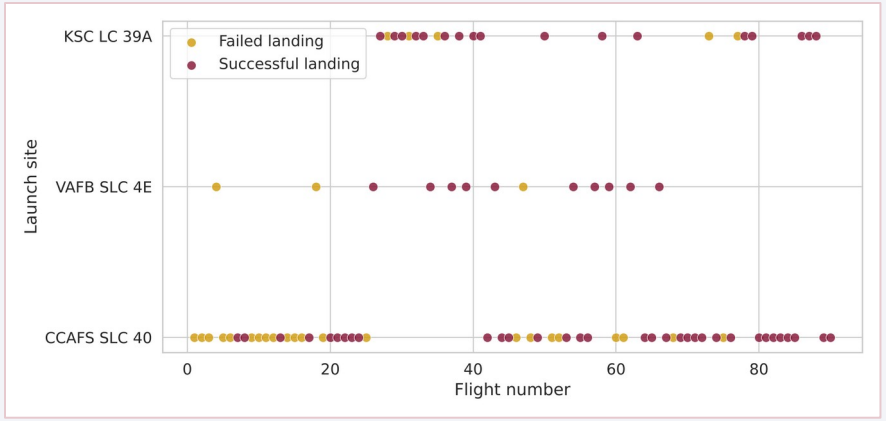
Model	Hyperparameters searched	Best parameters found
Logistic Regression	C: 0.01, 0.1, 1 · penalty l2 · solver lbfgs	C = 0.01
SVM	kernel: linear, rbf, poly, sigmoid · C, gamma: 10^-3 to 10^3	kernel sigmoid, C = 1, gamma = 0.032
Decision Tree	criterion, splitter, max_depth 2–18, max_features, min_samples	gini, depth 6, sqrt, leaf 4, split 5
KNN	n_neighbors 1–10 · algorithm · p: 1, 2	n_neighbors 10, p = 1

Each model was tuned on training data only, then scored once on the 18 held-out launches.

GitHub (machine learning): https://github.com/RoShawn-F/spacex-falcon9-landing-prediction/blob/main/SpaceX-Machine-Learning-Prediction-Part-5-v1.ipynb

EDA with Visualization: Results (1 of 3)

Flight Number vs. Launch Site (scatter plot)

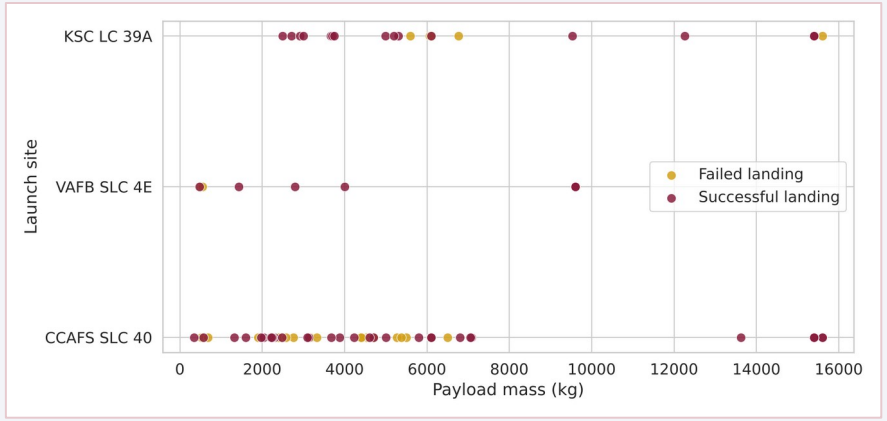


Why this chart: a scatter plot shows every launch at once, so we can see whether outcomes at each site improve as SpaceX gains experience (flight number).

Evidence: at CCAFS SLC 40, flights 1–20 landed 28% of the time (5 of 18) versus 100% for flights 81–90 (7 of 7). Overall: CCAFS SLC 40 60% (33 of 55), KSC LC 39A 77% (17 of 22), VAFB SLC 4E 77% (10 of 13).

Finding: success rises with flight number at every site; experience is the strongest pattern in the data.

Payload Mass vs. Launch Site (scatter plot)



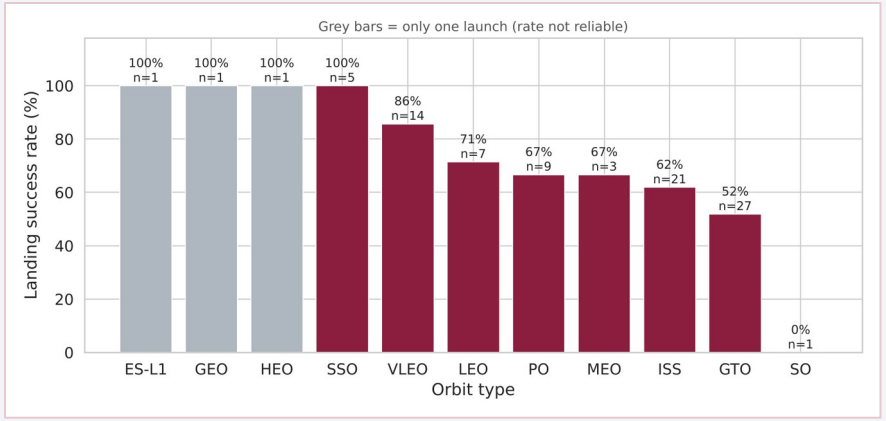
Why this chart: a scatter plot reveals whether heavier payloads are linked to failures at each launch site.

Evidence: landing rate by payload: under 4,000 kg 67% (26 of 39); 4,000–8,000 kg 50% (14 of 28); over 8,000 kg 87% (20 of 23). VAFB SLC 4E never launched more than 9,600 kg.

Finding: heavy payloads did not reduce success; they were flown later, when landings were more reliable.

EDA with Visualization: Results (2 of 3)

Success Rate vs. Orbit Type (bar chart)

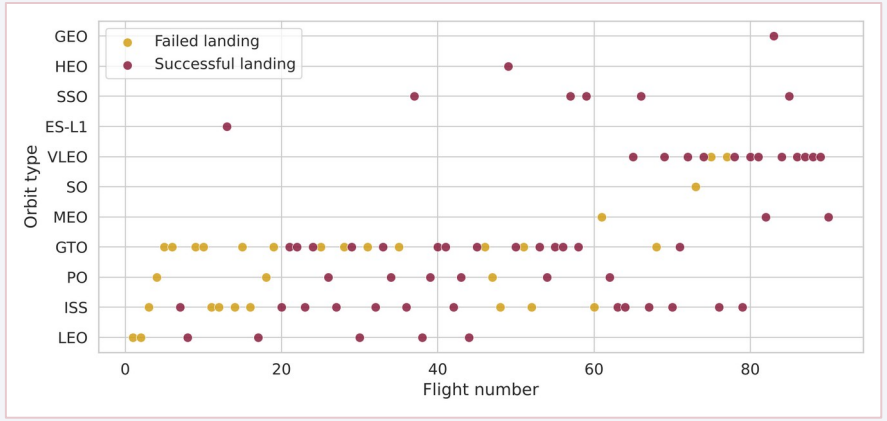


Why this chart: a bar chart is the clearest way to compare one rate (landing success) across categories (orbit types).

Evidence: SSO 5 of 5 (100%), VLEO 12 of 14 (86%), LEO 5 of 7 (71%), PO 6 of 9 (67%), MEO 2 of 3 (67%), ISS 13 of 21 (62%), GTO 14 of 27 (52%), SO 0 of 1. ES-L1, GEO and HEO: 1 of 1 each.

Finding: SSO and VLEO are reliable; GTO, the most common orbit, is the hardest. Single-launch orbits (grey) are not reliable evidence.

Flight Number vs. Orbit Type (scatter plot)



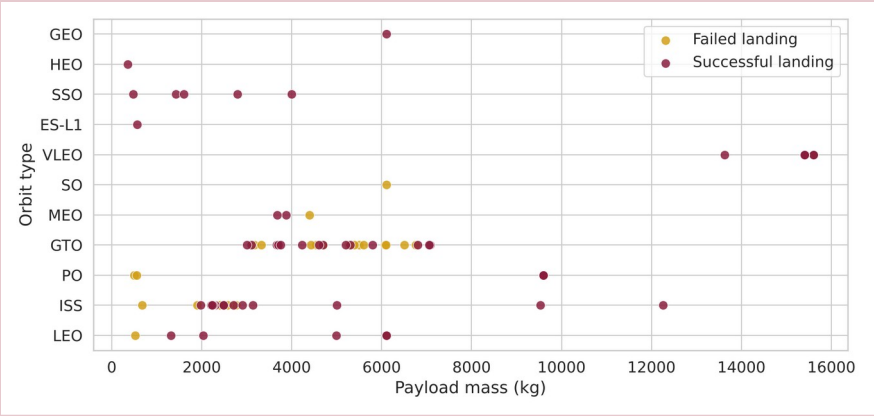
Why this chart: a scatter plot shows whether experience improves outcomes within each orbit.

Evidence: comparing the earlier and later half of each orbit's launches: LEO 33% → 100%, GTO 31% → 71%, ISS 50% → 73%, PO 50% → 80%.

Finding: success improves with experience in every major orbit, including GTO.

EDA with Visualization: Results (3 of 3)

Payload Mass vs. Orbit Type (scatter plot)

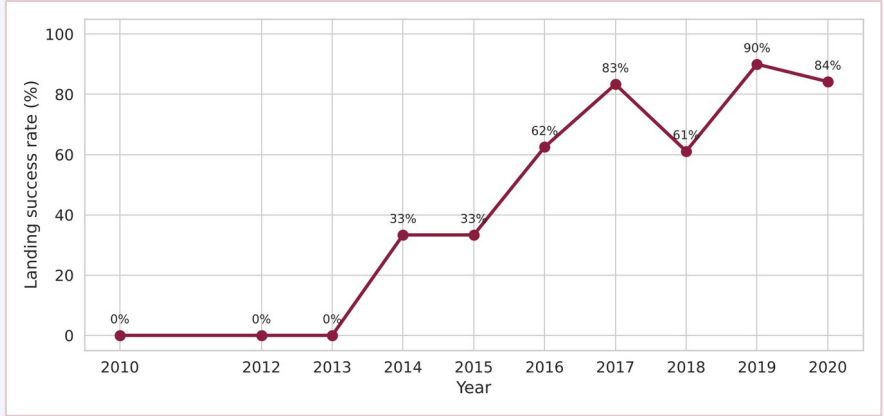


Why this chart: a scatter plot shows how payload weight relates to landing outcome for each orbit.

Evidence: payloads over 4,000 kg landed in PO (6 of 7), ISS (3 of 3) and LEO (3 of 4); VLEO missions (13,620–15,600 kg) landed 12 of 14. GTO: 5 of 7 at 4,000 kg or less, but 9 of 20 above 4,000 kg.

Finding: heavy payloads land well in PO, LEO, ISS and VLEO; GTO missions above 4,000 kg are the riskiest (45%).

Launch Success Yearly Trend (line chart)



Why this chart: a line chart is the standard way to show how a value changes over time.

Evidence: 2010–2013: 0 of 5 landed; 2014: 33%; 2015: 33%; 2016: 62%; 2017: 83%; 2018: 61%; 2019: 90%; 2020: 84%.

Finding: landing success rose from 0% to around 85–90% in seven years, with a dip in 2018.

EDA with SQL: Queries and Results (Tasks 1–4)

Task	SQL query	Result
1. Unique launch sites	`SELECT DISTINCT Launch_Site FROM SPACEXTABLE;`	CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40 (4 sites)
2. Sites beginning with 'CCA'	`SELECT * FROM SPACEXTABLE` `WHERE Launch_Site LIKE 'CCA%' LIMIT 5;`	5 CCAFS LC-40 launches (2010–2013), all F9 v1.0, no landings
3. Total payload, NASA (CRS)	`SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTABLE` `WHERE Customer = 'NASA (CRS)';`	45,596 kg
4. Average payload, F9 v1.1	`SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTABLE` `WHERE Booster_Version = 'F9 v1.1';`	2,928.4 kg

EDA with SQL: Queries and Results (Tasks 5–7)

Task	SQL query	Result
5. First ground-pad landing	`SELECT MIN(Date) FROM SPACEXTABLE` `WHERE Landing_Outcome = 'Success (ground pad)';`	2015-12-22

6. Drone-ship success, 4,000–6,000 kg	`SELECT Booster_Version FROM SPACEXTABLE`
	`WHERE Landing_Outcome = 'Success (drone ship)'`
	`AND PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ < 6000;`
		F9 FT B1022, F9 FT B1026, F9 FT B1021.2, F9 FT B1031.2

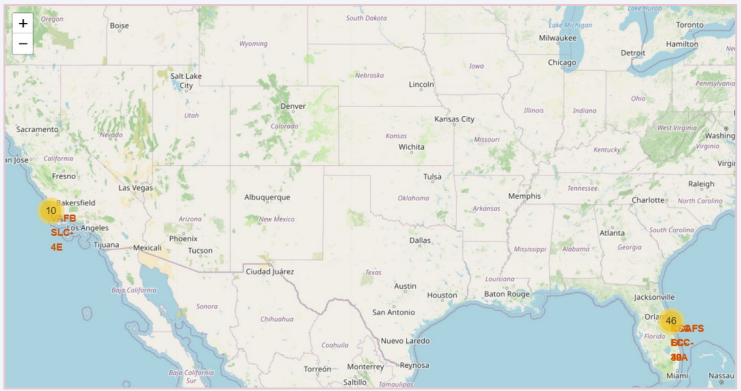
7. Mission outcomes	`SELECT Mission_Outcome, COUNT(*) FROM SPACEXTABLE` `GROUP BY Mission_Outcome;`	Success 100 (98 + 1 + 1 payload status unclear); Failure (in flight) 1

EDA with SQL: Queries and Results (Tasks 8–10)

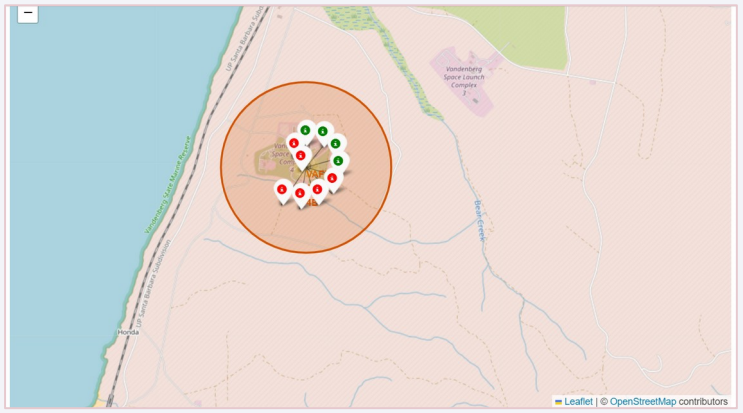
Task	SQL query	Result
8. Boosters with maximum payload	`SELECT DISTINCT Booster_Version FROM SPACEXTABLE WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE);`	12 F9 B5 boosters at 15,600 kg (B1048.4, B1049.4, B1051.3, B1056.4, B1048.5, B1051.4, B1049.5, B1060.2, B1058.3, B1051.6, B1060.3, B1049.7)
9. 2015 drone-ship failures	`SELECT substr(Date,6,2), Landing_Outcome, Booster_Version, Launch_Site FROM SPACEXTABLE WHERE Landing_Outcome = 'Failure (drone ship)' AND substr(Date,0,5) = '2015';`	Month 01: F9 v1.1 B1012; Month 04: F9 v1.1 B1015 (both CCAFS LC-40)
10. Landing outcomes ranked, 2010-06-04 to 2017-03-20	`SELECT Landing_Outcome, COUNT(*) AS Total FROM SPACEXTABLE WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY Total DESC;`	No attempt 10 · Success (drone ship) 5 · Failure (drone ship) 5 · Success (ground pad) 3 · Controlled (ocean) 3 · Uncontrolled (ocean) 2 · Failure (parachute) 2 · Precluded (drone ship) 1

Folium Maps: Launch Sites, Outcomes and Proximities

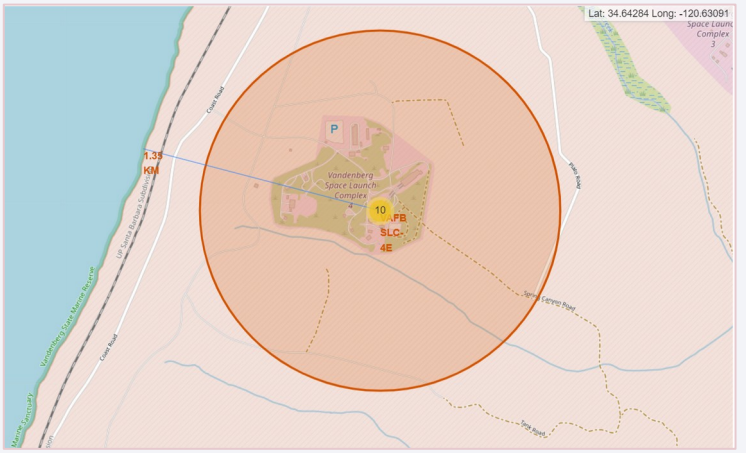
1. All launch sites



2. Launch outcomes (VAFB SLC-4E)



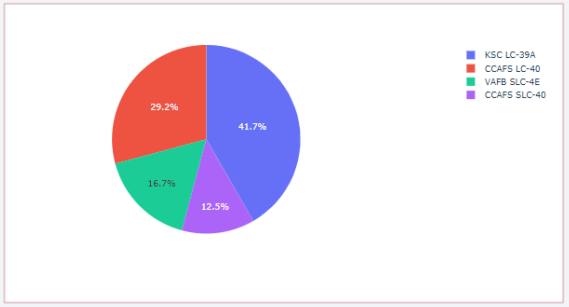
3. Distance to coastline (1.35 km)



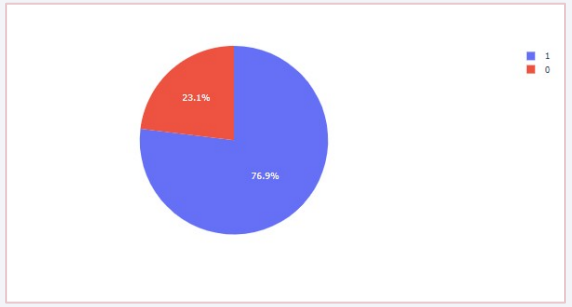
- Sites: 1 in California (VAFB SLC-4E, 10 launches) and 3 in Florida (46 launches); all are on the coast and as far south as practical
- Outcomes: green = landed, red = failed. VAFB SLC-4E landed 4 of 10; KSC LC-39A is best overall (10 of 13)
- Proximities from VAFB SLC-4E: coastline 1.35 km, railway 1.25 km, highway 1.15 km, nearest city (Santa Maria) 38.98 km
- Sites are close to the coast and transport for safety and logistics, but kept far from cities to protect residents

Plotly Dash: Success by Site and Payload vs. Outcome

1. Successful launches, all sites



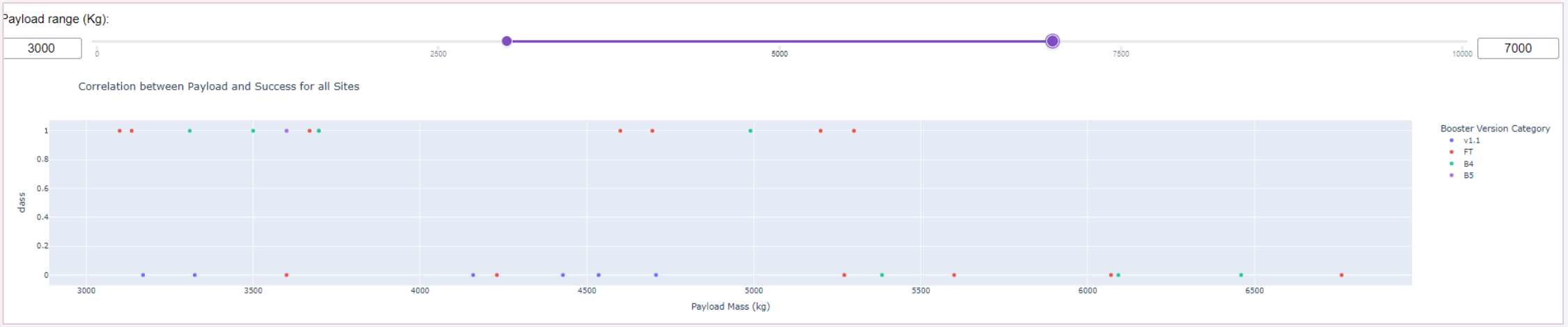
2. KSC LC-39A (highest ratio)



Key findings

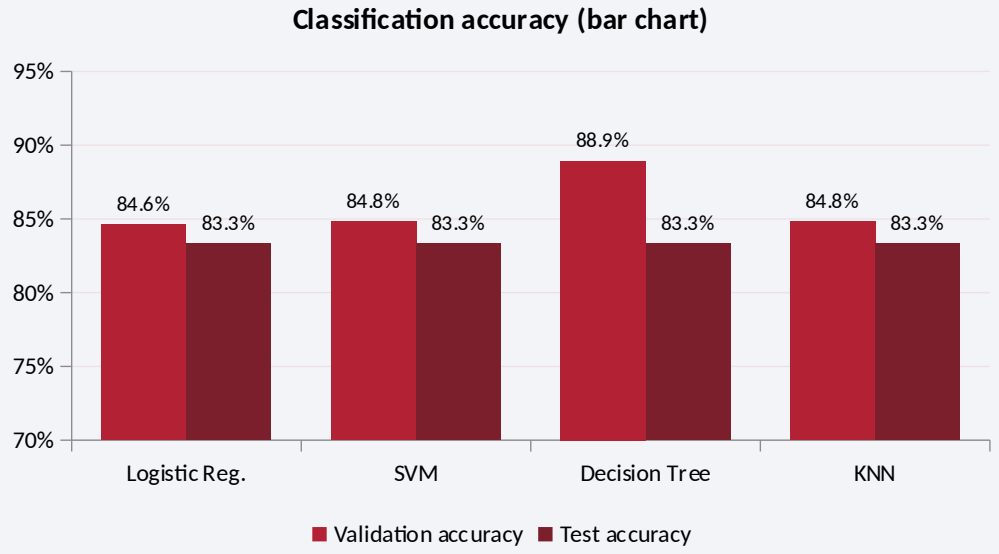
- KSC LC-39A has the largest share of successful landings (41.7%), then CCAFS LC-40 (29.2%), VAFB SLC-4E (16.7%), CCAFS SLC-40 (12.5%)
- KSC LC-39A also has the highest success ratio: 76.9% landed (10 of 13)
- Payload 3,000–7,000 kg: 13 of 28 landed; FT 8 of 14, B4 4 of 7, v1.1 0 of 6; all 5 launches above 5,500 kg failed

3. Payload vs. launch outcome scatter, all sites, range slider set to 3,000–7,000 kg (color = booster version)

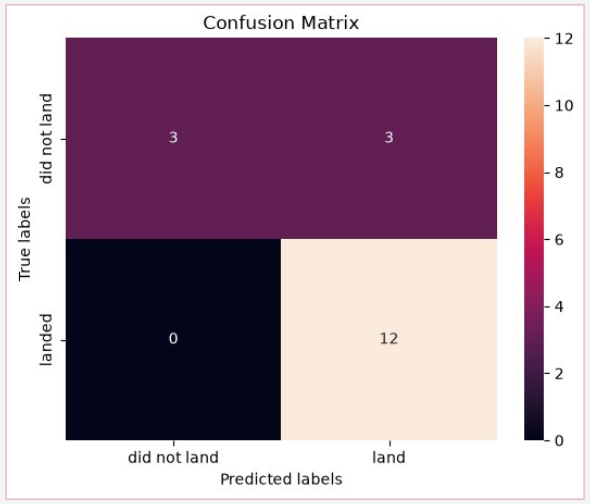


Predictive Analysis Results: Comparing 4 Classifiers

Model	Best hyperparameters (GridSearchCV, cv=10)	Validation accuracy	Test accuracy
Logistic Regression	C = 0.01, penalty l2, solver lbfgs	84.6%	83.3%
Support Vector Machine	kernel sigmoid, C = 1, gamma = 0.032	84.8%	83.3%
Decision Tree (best)	gini, max_depth 6, max_features sqrt, min_samples_leaf 4, min_samples_split 5	88.9%	83.3%
K-Nearest Neighbors	n_neighbors 10, p = 1, algorithm auto	84.8%	83.3%



Confusion matrix: Decision Tree



Evaluation

- Best model: Decision Tree, highest validation accuracy (88.9%)
- All 4 models: 83.3% test accuracy (15 of 18)
- 12 true positives, 3 true negatives, 3 false positives, 0 false negatives
- Weakness: false positives (predicts a landing that fails)

Conclusions

What we found

1. Experience is the strongest driver of success. Landing success rose from 0% (2010–2013) to 90% in 2019. Within every major orbit, later launches succeed far more often than earlier ones (for example, LEO 33% → 100%, GTO 31% → 71%).

2. Launch site matters. KSC LC-39A has the best success ratio (77%) and the largest share of successful landings (41.7%). All sites are on the coast, about 1–1.4 km from railways and highways, and far from cities (Santa Maria is 39 km from VAFB SLC-4E).

3. Payload and booster version both matter. Heavy payloads (over 8,000 kg) landed 87% of the time. Newer FT and B4 boosters land far more often than v1.0 and v1.1 boosters, which landed only 1 of 20 times.

4. Orbit type affects risk. SSO (100%) and VLEO (86%) are reliable; GTO, the most common orbit, is the hardest at 52%, and below 50% for payloads over 4,000 kg.

5. Landing success is predictable. A tuned Decision Tree reached 88.9% validation accuracy and 83.3% test accuracy (15 of 18). It never missed a real landing but made 3 false-positive predictions.

Implications for Space Y

- Use the model to estimate the chance a Falcon 9 first stage will be reused, and therefore SpaceX's real cost per launch, before bidding.
- Expect the lowest SpaceX costs on SSO, VLEO and LEO missions from KSC LC-39A, and higher costs on heavy GTO missions.
- Treat model predictions of a landing with caution: tune the decision threshold to reduce false positives so bids are not set too low.
- Retrain regularly: SpaceX's success rate is still rising, so recent launches matter most.

Innovative Insights

Landing is a learned skill

- Grouping launches into blocks of 20 flights shows success rising 25% → 80% → 70% → 75% → 100%. A bid model should weight recent launches more heavily.

Beware of 100% success rates

- ES-L1, GEO and HEO show 100% success from a single launch each. I flagged these in grey on the orbit chart so they are not mistaken for proven results.

The test set is too small to pick a winner

- With 18 test launches, each error moves accuracy by 5.6 points, which is why all four models tie at 83.3%. More data or repeated cross-validation would separate them.

Translate errors into dollars

- A false positive assumes a reusable booster, so SpaceX looks cheaper than it really is. Space Y should tune the decision threshold to reduce false positives before bidding.

Takeaway Resilient pipeline: when the live SpaceX API failed, the project switched to IBM's saved copy without changing the analysis.

Thank you!

